

Hawkeye EQMS — Process-Flow Analysis

Per-module lifecycle, persona × action × API × screen mapping, AI integration. Source-of-truth document for the test plan and demo scripts.

Document version	v2.1
Revised on	2026-04-24
Revised by	Hawkeye Engineering
Revision note	Extends v2.0 (5 modules) to v2.1 (14 modules) — adds Training, Change Control, Complaints, Internal Audit, Batch Records, Equipment/Calibration, Supplier Pre-Qualification, Audit Request/RFQ, Design Control.
Modules covered	14 (Deviation / Non-Conformance · CAPA — Corrective & Preventive Action (v2) · Document Control (SOP / WI / Policy / Spec) · Risk Register (FMEA) · Management Review (MRM) · Training Records · Change Control · Complaint Management · Internal Audit Execution (8-phase lifecycle) · Batch / Manufacturing Records · Equipment / Calibration · Supplier Pre-Qualification · Audit Request / RFQ (marketplace audit flow) · Design Control (medical device))
Total lifecycle states	111
Total legal transitions	95
AI assists wired in	20

Contents

- 1. Deviation / Non-Conformance
- 2. CAPA — Corrective & Preventive Action (v2)
- 3. Document Control (SOP / WI / Policy / Spec)
- 4. Risk Register (FMEA)
- 5. Management Review (MRM)
- 6. Training Records
- 7. Change Control
- 8. Complaint Management
- 9. Internal Audit Execution (8-phase lifecycle)
- 10. Batch / Manufacturing Records
- 11. Equipment / Calibration
- 12. Supplier Pre-Qualification
- 13. Audit Request / RFQ (marketplace audit flow)
- 14. Design Control (medical device)
- 15. Cross-cutting infrastructure

1. Deviation / Non-Conformance

Compliance	ISO 9001:2015 §10.2 · 21 CFR 211.192 · ICH Q10 §3.2.3
Purpose	Capture, investigate, disposition and close any planned or unplanned deviation from a specification, SOP, batch record, or quality system requirement. Drive CAPA when systemic.
Model file	backend/src/models/DeviationModel.js
Collection	deviations
Numbering	DEV-YYYY-NNNN
Classification	CRITICALMAJORMINOR

LIFECYCLE STATES

State	Primary owner	Description
REPORTED	QA Specialist (admin)	Initial intake; raised by detector via 'Report Deviation' dialog. Captures title, description, classification, type, category, area, processStep, productName, batchNumbers, immediateActions.
UNDER_ASSESSMENT	QA Specialist / Head of QA	Impact assessment under way: product quality, patient safety, batch disposition, regulatory impact.
UNDER_INVESTIGATION	QA Specialist / SME	Root-cause analysis. Method may be 5-Why / Fishbone / FMEA / FTA. AI 5-Why scaffolder available.
PENDING_DISPOSITION	Head of QA	RCA complete; awaiting batch disposition decision (RELEASE / REJECT / REWORK / REPROCESS / QUARANTINE).
PENDING_CAPA_DECISION	Head of QA	Disposition done; deciding whether systemic CAPA is required.
CAPA_REQUIRED	QA / CAPA Owner	CAPA required and being created (auto via cross-module service or linked manually).
PENDING_CLOSURE	Head of QA / VP Quality	All actions closed; final review before formal closure.
CLOSED	VP Quality (tenant_admin)	Terminal. closedBy + closedAt + closureNotes captured. Triggers e-sig if configured.

State	Primary owner	Description
CANCELLED	Head of QA	Terminal. Raised in error; cancel reason captured.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
REPORTED	UNDER_ASSESSMENT	Assess	POST /api/deviations/:id/assess	productQualityImpact, patientSafetyImpact, batchDisposition, regulatoryImpact	EDITOR_ROLES (admin/buyer/auditor/tenant_admin/insp)
REPORTED UNDER_ASSESSMENT	UNDER_INVESTIGATION	Investigate	POST /api/deviations/:id/investigate	method, summary, rootCause, rootCauseCategory	EDITOR_ROLES
UNDER_INVESTIGATION	PENDING_DISPOSITION	Investigate (sets rootCause)	POST /api/deviations/:id/investigate (with rootCause)	rootCause set	EDITOR_ROLES
PENDING_DISPOSITION	PENDING_CAPA_DECISION	Dispose	POST /api/deviations/:id/dispose	decision, justification	EDITOR_ROLES
PENDING_CAPA_DECISION	CAPA_REQUIRED PENDING_CLOSURE	(decision drawer)	POST /api/deviations/:id/capa-decision	capaRequired (bool); optional autoCreateCapa	EDITOR_ROLES
CAPA_REQUIRED PENDING_CLOSURE	CLOSED	Close	POST /api/deviations/:id/close	closureNotes	EDITOR_ROLES
(any)	CANCELLED	Cancel	DELETE /api/deviations/:id (soft)	n/a	EDITOR_ROLES

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
5-Why Scaffolder (Wave 1)	POST /api/ai/deviation/scaffold-five-why	UNDER_INVESTIGATION	View+AI drawer · 'Scaffold 5-why with AI' popover	Generates a 5-level why chain with citations + 6M (Man/Machine/Method/Material/Measurement/Environment) categorisation. Free Gemini.
CAPA RCA Drafter (Wave 1)	POST /api/ai/capa/draft-rca	UNDER_INVESTIGATION → CAPA_REQUIRED	View+AI drawer · 'Draft CAPA RCA' drawer	Drafts severity + RCA + corrective/preventive actions + effectiveness plan. User edits + e-signs before saving.
Predictive CAPA badge (Wave 3)	POST /api/ai/predict/capa-outcome	PENDING_CAPA_DECISION	View+AI drawer (auto-renders)	Heuristic prediction P(on-time) + P(effective) + top contributing factors.
Signal cluster detector (Wave 3)	GET /api/ai/signals?status=open	all states (post-create)	Head-of-QA dashboard	Z-score trend detection on equipment / material / process clusters.

SCREENS

- /nonconformance — List + report + lifecycle actions
- /nonconformance (View+AI drawer) — Detail + AI 5-Why + Draft CAPA RCA + Predictive badge

CLOSURE CONDITIONS

CLOSED requires (a) status set via /close, (b) closureNotes captured, (c) closedBy + closedAt set. If CAPA was required, the linked CAPAs must also reach CLOSED_EFFECTIVE before the deviation is permanently closed.

2. CAPA — Corrective & Preventive Action (v2)

Compliance	ISO 9001:2015 §10.2 · 21 CFR 211.192 · ICH Q10 §3.2.2
Purpose	Formalise root-cause investigation + corrective + preventive actions for a quality issue. Track effectiveness to closure. Enforce 5-stage approval gates (Triage / RCA / Action Plan / Effectiveness / Closure).
Model file	backend/src/models/capaV2Models.js
Collection	capa-v2 + 16 child collections
Numbering	CAPA-YYYY-NNNN

LIFECYCLE STATES

State	Primary owner	Description
DRAFT_CANDIDATE	QA / Auditor	Candidate created from an audit finding or manual draft.
INTAKE_DRAFT	Submitter	Intake form being filled (issue statement, scope, regulatory exposure).

State	Primary owner	Description
UNDER_TRIAGE	Triage Reviewer (auditor/admin)	Triage decision pending: FORMAL_CAPA / CORRECTION_ONLY / NO_CAPA.
TRIAGE_NO_CAPA	Triage Reviewer	Terminal — no CAPA needed; rationale logged.
CORRECTION_ONLY	CAPA Owner	Quick fix only; no formal investigation.
CAPA_OPEN	CAPA Owner	Formal CAPA opened; ownerUserId + dueDate required.
INVESTIGATION_IN_PROGRESS	Investigator	Investigation scope + activities being recorded.
RCA_PENDING_APPROVAL	Approver (auditor/admin)	Root-cause analysis submitted; awaiting approval.
ACTION_PLAN_PENDING_APPROVAL	Approver	Corrective/preventive plan submitted; awaiting approval.
ACTION_PLAN_APPROVED	CAPA Owner	Plan approved; ready to execute action items.
IN_IMPLEMENTATION	Action-item Owners	Action items being worked. Evidence attached as proof.
AWAITING_EFFECTIVENESS_CHECK	Effectiveness Reviewer	Implementation done; awaiting effectiveness evaluation.
EFFECTIVENESS_REVIEW_IN_PROGRESS	Effectiveness Reviewer	Evaluating whether actions achieved their goal (PASS/FAIL).
CLOSED_EFFECTIVE	VP Quality	Terminal. Effective; closure approved.
CLOSED_INEFFECTIVE	VP Quality	Terminal. Ineffective; reopen or supersede needed.
REOPENED	QA / Reviewer	Reopened after recurrence or failed effectiveness.
CANCELLED	Approver	Terminal.
SUPERSEDED	VP Quality	Terminal — replaced by another CAPA.
MERGED	VP Quality	Terminal — merged into another CAPA.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	AI
DRAFT_CANDIDATE	INTAKE_DRAFT UNDER_TRIAGE MERGED CANCELLED	(workspace tabs)	POST /api/capa-v2/intakes (or merge endpoint)	intake fields	m
INTAKE_DRAFT	UNDER_TRIAGE CANCELLED	Submit Intake	POST /api/capa-v2/intakes/:id/submit	all intake fields	m
UNDER_TRIAGE	TRIAGE_NO_CAPA CORRECTION_ONLY CAPA_OPEN	Triage decision	POST /api/capa-v2/triage/:id/decision	decision (NO_CAPA/CORRECTION/FORMAL_CAPA), rationale	tri (ai)
CAPA_OPEN	INVESTIGATION_IN_PROGRESS	Open CAPA	POST /api/capa-v2/capas (then PUT investigation)	ownerUserId, dueDate	m
INVESTIGATION_IN_PROGRESS	RCA_PENDING_APPROVAL	Submit RCA	PUT /api/capa-v2/capas/:id/root-cause	rootCauseStatement, rcaMethod	m
RCA_PENDING_APPROVAL	ACTION_PLAN_PENDING_APPROVAL	Approve RCA	POST /api/capa-v2/capas/:id/approvals (stage=RCA, decision=APPROVED)	approver decision	ap (ai)
ACTION_PLAN_PENDING_APPROVAL	ACTION_PLAN_APPROVED	Approve Plan	POST /api/capa-v2/capas/:id/approvals (stage=ACTION_PLAN, decision=APPROVED)	approver decision	ap (ai)
ACTION_PLAN_APPROVED	IN_IMPLEMENTATION	Start Action	PATCH /api/capa-v2/action-items/:id/status (start any item)	≥ 1 action item moved to IN_PROGRESS	m
IN_IMPLEMENTATION	AWAITING_EFFECTIVENESS_CHECK	Mark Complete	POST /api/capa-v2/capas/:id/implementation-evidence (last action complete)	all action items COMPLETED + evidence	m
AWAITING_EFFECTIVENESS_CHECK	EFFECTIVENESS_REVIEW_IN_PROGRESS	Begin EC	PUT /api/capa-v2/capas/:id/effectiveness (start)	review window dates	m
EFFECTIVENESS_REVIEW_IN_PROGRESS	CLOSED_EFFECTIVE CLOSED_INEFFECTIVE	Close CAPA	POST /api/capa-v2/capas/:id/close	closureOutcome (EFFECTIVE/INEFFECTIVE), closedAt	ap (ai)
CLOSED_EFFECTIVE CLOSED_INEFFECTIVE	REOPENED SUPERSEDED MERGED	Reopen	POST /api/capa-v2/capas/:id/reopen (or supersede)	reason	ap (ai)

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
CAPA RCA Drafter	POST /api/ai/capa/draft-rca	INVESTIGATION_IN_PROGRESS	From deviation View+AI drawer or CAPA workspace	Drafts severity + RCA + actions; user edits + e-signs.
Predictive CAPA Outcome (Wave 3)	POST /api/ai/predict/capa-outcome	CAPA_OPEN onwards	Predictive badge on CAPA row	P(on-time) + P(effective) heuristic.
Active Learning Loop	POST /api/ai/active-learning/adjustments	CLOSED_EFFECTIVE/CLOSED_INEFFECTIVE	Drift dashboard	Refines retrieval weights based on accept/reject outcomes.

SCREENS

- /buyer/capas — CAPA workspace (admin/buyer view)
- /auditor/capas — CAPA workspace (auditor view, scoped to assigned)

CLOSURE CONDITIONS

CLOSED_EFFECTIVE requires: all action items COMPLETED, ≥1 implementation evidence, effectiveness check PASS, closure approval (stage=CLOSURE) by approvalRoles user. CLOSED_INEFFECTIVE permitted but triggers reopen workflow.

3. Document Control (SOP / WI / Policy / Spec)

Compliance	ISO 9001:2015 §7.5 · 21 CFR 211.180 · 21 CFR Part 11 §11.10 (e-signature)
Purpose	Author, review, approve, publish, supersede and withdraw controlled documents. Multi-step serial approval. Triggers training auto-assignment when published if requiresTrainingOnUpdate=true.
Model file	backend/src/models/DocumentControlModel.js
Collection	document-controls
Numbering	Sequential per tenant; user-set documentNumber

LIFECYCLE STATES

State	Primary owner	Description
DRAFT	Doc Author	Initial draft; freely editable.
UNDER_REVIEW	Reviewers (per step)	approvalSteps[] active; serial review per step.
APPROVED	Doc Control	All steps APPROVED/DELEGATED; awaiting publish.
EFFECTIVE	(All users)	Live; effectiveDate set. Supersede or withdraw permitted.
SUPERSEDED	Doc Control	Terminal; supersededById points to new revision.
WITHDRAWN	Doc Control	Terminal; withdrawalReason captured.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
DRAFT	UNDER_REVIEW	Submit for Review	POST /api/document-control/:id/submit-for-review	reviewers[] (each with role; userId optional)	authenticated tenant user
UNDER_REVIEW	DRAFT (rejected) UNDER_REVIEW (next step) APPROVED (all approved)	Approve / Reject	POST /api/document-control/:id/approve	stepOrder, decision (APPROVED/REJECTED/DELEGATED), comments	authenticated; step.approverId enforced
APPROVED	EFFECTIVE	Publish	POST /api/document-control/:id/publish	effectiveDate (defaults to today); triggers training auto-assign	doc admin
EFFECTIVE	SUPERSEDED	Supersede	POST /api/document-control/:id/supersede	new doc payload (auto-creates new revision)	doc admin
EFFECTIVE	WITHDRAWN	Withdraw	POST /api/document-control/:id/withdraw	withdrawalReason	doc admin

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
Training Auto-Assign Agent (Wave 2)	POST /api/ai/training/auto-assign-on-sop-revision	EFFECTIVE (publish)	Auto-fired on publish if requiresTrainingOnUpdate=true	Identifies affected roles + creates 'read-and-understood' training records with grace-period dueDate. Optionally drafts an LLM knowledge-check question.

SCREENS

- /document-control — Register + state-action buttons + 'Effective' indicator

CLOSURE CONDITIONS

EFFECTIVE is the operational state. Permanent closure occurs at SUPERSEDED (replaced by new revision) or WITHDRAWN (taken out of service). versionMajor auto-increments on supersede.

4. Risk Register (FMEA)

Compliance	ISO 9001:2015 §6.1 · ICH Q9 (R1) Quality Risk Management
Purpose	Capture process-step risks (failure mode → effect → cause) with FMEA scoring (S × O × D = RPN). Add mitigations until residual risk acceptable. Tie to source (audit, deviation, complaint, CAPA, regulatory).
Model file	backend/src/models/RiskItemModel.js
Collection	risk-items
Numbering	user-defined / N/A

FMEA RPN BANDS

LOW	RPN < 60
MEDIUM	RPN 60 – 124
HIGH	RPN 125 – 199
CRITICAL	RPN ≥ 200

LIFECYCLE STATES

State	Primary owner	Description
OPEN	Risk Owner	Initial; no mitigations completed.
MITIGATED	Risk Owner	≥ 1 mitigation COMPLETED; residual S/O/D entered; residual RPN computed.
ACCEPTED	VP Quality	Residual risk acceptable; sign-off captured.
CLOSED	VP Quality	Terminal; full audit trail verified.
TRANSFERRED	VP Quality	Terminal; ownership moved to another tenant/site.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
OPEN	OPEN (with mitigation appended)	Add Mitigation	POST /api/risk-items/:id/mitigate	action, owner, dueDate	tenant authenticated
OPEN	MITIGATED	Mark Mitigated	PUT /api/risk-items/:id (status=MITIGATED + residual S/O/D)	≥ 1 mitigation COMPLETED; residual scores	risk owner
MITIGATED	ACCEPTED	Accept Residual	PUT /api/risk-items/:id (status=ACCEPTED)	approver sign-off	tenant_admin
ACCEPTED	CLOSED	Close	PUT /api/risk-items/:id (status=CLOSED)	closure note	tenant_admin
(any non-terminal)	TRANSFERRED	Transfer	PUT /api/risk-items/:id (status=TRANSFERRED)	transfer destination	tenant_admin

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
Risk Scenario Brainstormer (Wave 2)	POST /api/ai/risk/brainstorm-scenarios	OPEN (creation)	Risk Register · 'Brainstorm with AI' button	Given a process description + product class, AI proposes failure modes + effects + causes seeded into draft risk items.

SCREENS

- /risk-register — FMEA register with RPN/Band columns + Log Risk dialog

CLOSURE CONDITIONS

CLOSED requires (a) all mitigations COMPLETED, (b) residual RPN acceptable per tenant policy, (c) tenant_admin approval. Residual RPN auto-computed on save when residualSeverity + residualOccurrence + residualDetectability all set.

5. Management Review (MRM)

Compliance	ISO 9001:2015 §9.3 · 21 CFR Part 11 (e-signed minutes)
Purpose	Periodic top-management review of QMS performance: audit results, CAPA status, deviations, complaints, supplier performance, resource adequacy, improvement opportunities, decisions and action items.
Model file	backend/src/models/ManagementReviewModel.js

Collection	management-reviews
Numbering	MR-YYYY-NNNN

LIFECYCLE STATES

State	Primary owner	Description
PLANNED	Chair (VP Quality)	Scheduled; plannedDate + reviewType + chair + attendees set.
IN_PROGRESS	Chair / Minutes Taker	Meeting under way; inputs[] being captured (manually or via AI populator).
COMPLETED	Chair + Approver	Terminal; outputs (qmsAdequacy / resourceDecisions / improvementOpportunities / actionItems) captured + e-signed.
CANCELLED	Chair	Terminal.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
PLANNED	IN_PROGRESS	Start Meeting	PUT /api/management-reviews/:id (status=IN_PROGRESS, actualDate)	actualDate	tenant authenticated
IN_PROGRESS	COMPLETED	Complete + Sign	POST /api/management-reviews/:id/complete	qmsAdequacy, resourceDecisions, improvementOpportunities, actionItems[]	chair / tenant_admin
PLANNED IN_PROGRESS	CANCELLED	Cancel	DELETE /api/management-reviews/:id (soft) or PUT status=CANCELLED	reason	chair

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
MRM Input Populator (Wave 2)	POST /api/ai/mrm/populate-inputs	IN_PROGRESS	MRM detail · 'Auto-populate inputs' button	Aggregates last N days of audits + CAPAs + deviations + complaints → populates inputs[] with topic/summary/trend. Adds AI narrative.

SCREENS

- /management-review — Register + Schedule Review dialog + detail with inputs + action items

CLOSURE CONDITIONS

COMPLETED requires (a) inputs[] populated for all 9.3.2 topics, (b) qmsAdequacy decision (ADEQUATE/NEEDS_IMPROVEMENT/INADEQUATE), (c) resourceDecisions + improvementOpportunities recorded, (d) actionItems[] with owners + due dates, (e) approvedBy + approvedAt set (e-signature), (f) optionally minutesDocumentId linked.

6. Training Records

Compliance	ISO 9001:2015 §7.2 · 21 CFR 211.25 · 21 CFR 211.100 (training on SOPs)
Purpose	Track assigned, in-progress and completed training for every employee. Auto-assign 'Read and Understood' records when SOPs are published. Capture competency assessment + knowledge-check results for audit readiness.
Model file	backend/src/models/TrainingRecordModel.js
Collection	training-records
Numbering	n/a

LIFECYCLE STATES

State	Primary owner	Description
ASSIGNED	Trainer / auto-assign agent	Initial state; assigned with dueDate.
IN_PROGRESS	Trainee	Trainee has started but not completed.
COMPLETED	Trainee	Terminal. Completion captured with competencyLevel + assessment.
OVERDUE	(auto)	Auto-escalation when dueDate < today and not COMPLETED.
WAIVED	Trainer	Terminal. Waiver captured with waiverReason + waivedBy.
FAILED	Trainer	Terminal. Assessment failed; remedial action required.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
(create)	ASSIGNED	Assign Training	POST /api/training-records	traineeld, trainingCode, trainingTitle, dueDate	authenticated tenant user / Wave 2 agent
ASSIGNED	IN_PROGRESS	Start Training	PUT /api/training-records/:id	status=IN_PROGRESS	trainee

From	To	UI button	API	Required input	Allowed roles
ASSIGNED IN_PROGRESS	COMPLETED	Mark Complete	POST /api/training-records/:id/complete	competencyLevel (AWARE/COMPETENT/PROFICIENT/EXPERT); assessment (type, score, passed, notes)	trainee
ASSIGNED IN_PROGRESS	WAIVED	Waive	PUT /api/training-records/:id (status=WAIVED)	waiverReason, waivedBy	trainer
ASSIGNED IN_PROGRESS	FAILED	Mark Failed	PUT /api/training-records/:id (status=FAILED)	assessment (passed=false)	trainer

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
Training Auto-Assign Agent (Wave 2)	POST /api/ai/training/auto-assign-on-sop-revision	ASSIGNED (creation)	Auto-fired on SOP publish when requiresTrainingOnUpdate=true	Given an SOP revision, identifies affected roles/departments and creates read-and-understood training records. Optionally drafts a multiple-choice knowledge-check via LLM.

SCREENS

- /training — Assigned + completed list, +Assign Training dialog, status/type filters

CLOSURE CONDITIONS

COMPLETED requires competencyLevel + assessment (type, score, passed). An e-signature may be required per tenant policy (21 CFR Part 11).

7. Change Control

Compliance	ICH Q10 §3.2.3 · 21 CFR 211.100 · 21 CFR 820.30 (medical device) · EU GMP Annex 15
Purpose	Control + document any proposed change to an approved process, material, equipment, specification, or procedure. Drive impact assessment, multi-level approval, implementation, and effectiveness verification.
Model file	backend/src/models/ChangeControlModel.js
Collection	change_controls
Numbering	user-defined

LIFECYCLE STATES

State	Primary owner	Description
DRAFT	Change originator	Draft change request; freely editable.
SUBMITTED	Change manager	Submitted for initial screening + assignment.
IMPACT_ASSESSMENT	Cross-functional SMEs	Regulatory + quality + technical impact captured (AI classifier available).
UNDER_REVIEW	Reviewers (approvalSteps[])	Multi-step serial approval per approvalSteps[] array.
APPROVED	Change manager	All approval steps APPROVED; ready to implement.
IMPLEMENTATION	Implementation owners	Change is being executed per the approved plan.
VERIFICATION	QA verifier	Effectiveness verification under way.
CLOSED	QA	Terminal. Effectiveness confirmed; change closed.
REJECTED	Reviewer	Terminal. Reject reason captured.
CANCELLED	Originator / Change mgr	Terminal.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
(create)	DRAFT	New Change Request	POST /api/universal/change-controls	title, description, changeType, riskLevel	buyer/supplier/auditor/admin/tenant_admin (createRoles)
DRAFT	SUBMITTED	Submit	PUT /api/universal/change-controls/:id (status=SUBMITTED)	full payload	createRoles
SUBMITTED	IMPACT_ASSESSMENT	Begin Impact Assessment	PUT /api/universal/change-controls/:id (status=IMPACT_ASSESSMENT)	initial triage	reviewRoles
IMPACT_ASSESSMENT	UNDER_REVIEW	Send for Review	PUT /api/universal/change-controls/:id (status=UNDER_REVIEW)	regulatoryImpact, impactAssessment fields	reviewRoles

From	To	UI button	API	Required input	Allowed roles
UNDER_REVIEW	APPROVED REJECTED	Approve / Reject	POST /api/universal/change-controls/:id/approval	stepOrder, decision (APPROVED/REJECTED/ABSTAINED), comments	auditor/admin/tenant_admin/reviewer/workflow_manage
APPROVED	IMPLEMENTATION	Start Implementation	PUT /api/universal/change-controls/:id (status=IMPLEMENTATION)	implementation plan	createRoles
IMPLEMENTATION	VERIFICATION	Begin Verification	POST /api/universal/change-controls/:id/verify-effectiveness	verificationNotes, effectivenessCheck, effective=false (pending)	createRoles
VERIFICATION	CLOSED	Confirm Effective + Close	POST /api/universal/change-controls/:id/verify-effectiveness (effective=true)	verificationNotes, effectivenessCheck, effective=true	createRoles

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
Regulatory Impact Classifier (Wave 2)	POST /api/ai/change-control/classify-impact	IMPACT_ASSESSMENT	Change-control detail · 'Classify regulatory impact' button	Given a change description + changeType + riskLevel + affected products/markets, classifies US (CBE-30 / PAS-NDA) + EU (Type IA / IB / II Variation) routes with reasoning.

SCREENS

- /change-controls — Register + New Change dialog + approval drawer

CLOSURE CONDITIONS

CLOSED requires (a) all approvalSteps APPROVED, (b) implementation evidence captured, (c) verify-effectiveness with effective=true, (d) optional e-sig on closure decision.

8. Complaint Management

Compliance	ISO 9001:2015 §10.2 · 21 CFR 211.198 · 21 CFR 820.198 (Complaint Files)
Purpose	Capture, investigate, and close customer complaints. Determine MDR reportability. Link to CAPA when systemic. Track regulatory reporting timelines.
Model file	backend/src/models/ComplaintModel.js
Collection	complaints
Numbering	auto (complaintNumber)

LIFECYCLE STATES

State	Primary owner	Description
OPEN	Complaint Intake	Initial intake; severity + type + source captured. isMedicalDeviceReport flag set.
UNDER_INVESTIGATION	Complaint Investigator	Investigation open; rootCause + investigationSummary being captured.
PENDING_CAPA	Complaint Investigator	Investigation indicates CAPA required; awaiting CAPA record creation + linking.
CAPA_IN_PROGRESS	CAPA Owner	Linked CAPA records being executed.
PENDING_CLOSURE	QA	All CAPAs closed; final review before formal closure.
CLOSED	QA	Terminal. closureNotes + correctiveAction + preventiveAction captured.
CANCELLED	QA	Terminal. Cancelled reason captured (OPEN-only via DELETE).

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
(create)	OPEN	+ Log Complaint	POST /api/complaints	title, complaintType, severity, source	authenticated tenant user
OPEN	UNDER_INVESTIGATION	Investigate	POST /api/complaints/:id/investigate	investigationSummary, rootCause, assignedTo	authenticated
UNDER_INVESTIGATION	PENDING_CAPA	Require CAPA	PUT /api/complaints/:id (status=PENDING_CAPA; linkedCAPAIds[])	linkedCAPAIds	authenticated
PENDING_CAPA	CAPA_IN_PROGRESS	(auto)	PUT /api/complaints/:id (status=CAPA_IN_PROGRESS)	CAPA opened	authenticated

From	To	UI button	API	Required input	Allowed roles
CAPA_IN_PROGRESS	PENDING_CLOSURE	(auto)	PUT /api/complaints/:id (status=PENDING_CLOSURE)	all CAPAs CLOSED_EFFECTIVE	authenticated
PENDING_CLOSURE	CLOSED	Close	POST /api/complaints/:id/close	closureNotes, correctiveAction, preventiveAction	authenticated
OPEN	CANCELLED	Cancel	DELETE /api/complaints/:id (OPEN-only)	—	authenticated

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
(planned) Complaint triage	(not yet wired)	OPEN	(future)	Wave 3 roadmap — pattern-match the complaint text against historical CAPAs + FDA MedWatch to pre-suggest severity, MDR reportability, and linked CAPAs.

SCREENS

- /complaint-manager — Register + Log Complaint dialog + investigation/closure drawers

CLOSURE CONDITIONS

CLOSED requires closureNotes; correctiveAction + preventiveAction captured. If isMedicalDeviceReport=true, MDR 3500A submission timeline is tracked separately (FDA eMDR).

9. Internal Audit Execution (8-phase lifecycle)

Compliance	ISO 19011 · ISO 9001:2015 §9.2 · 21 CFR 211.180 · ICH Q10 §3.2.4 · EU GMP Chapter 9
Purpose	Run a supplier or internal audit through 8 sequential phases — from intake through surveillance follow-up. Each phase owns specific artifacts (DRL, SMF, agenda, checklist, PDR, CAPA, closure certificate). Role gates enforce buyer/supplier/auditor collaboration.
Model file	backend/src/models/auditRequestsMasterModel.js
Collection	audit-requests-master
Numbering	HAWK-XXXXXXXXXX (global) + supplier-scoped

LIFECYCLE STATES

State	Primary owner	Description
INITIATED	Buyer	Buyer creates audit request. RFQ/intimation + scope captured.
PREP	Supplier	Supplier completes pre-audit questionnaire + Document Requirements List + Site Master File.
PLANNING	Auditor	Auditor builds scope + agenda + COI declaration.
EXECUTION	Auditor	On-site / remote audit under way. Opening meeting, observations, evidence capture.
FINDINGS	Auditor	Findings documented per ISO 19011 severity (Critical/Major/Minor/Observation). Preliminary Deficiency Report issued.
CAPA	Supplier → Buyer	Supplier submits CAPA plans + evidence; buyer reviews.
CLOSURE	Buyer	Final report approved; closure certificate signed by buyer/supplier/auditor/witness.
SURVEILLANCE	Buyer/Auditor	Follow-up audit scheduled per surveillance cadence.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
(create)	INITIATED	Request Audit	POST /api/audit-requests (buyer)	supplierOrgId, scope, auditType	buyer/tenant_admin
INITIATED	PREP	Accept Audit	POST /api/audit-requests/:id/supplier-decision (ACCEPTED)	supplierDecision, optional proposedDates	supplier
INITIATED	(rejected)	Propose / Reject	POST /api/audit-requests/:id/supplier-decision (REJECTED/PROPOSED)	supplierDecision, dispute reason	supplier
any	any (admin)	Force Transition	POST /api/audits/:id/phases/transition (override=true)	override flag (admin-only)	admin/tenant_admin
PREP	PLANNING	Advance to Planning	POST /api/audits/:id/phases/transition	prep artifacts complete	auditor/buyer
PLANNING	EXECUTION	Start Execution	POST /api/audits/:id/phases/transition	agenda + scope approved	auditor
EXECUTION	FINDINGS	Advance to Findings	POST /api/audits/:id/phases/transition	closing meeting complete	auditor

From	To	UI button	API	Required input	Allowed roles
FINDINGS	CAPA	Validate Findings	POST /api/audit-requests/:id/deficiency-validation	deficiencyValidation (ACCEPTED/PARTIALLY_ACCEPTED/DISPUTED) + disputeltems	supplier
CAPA	CLOSURE	Close Audit	POST /api/audits/:id/phases/transition	all linked CAPAs closed	buyer
CLOSURE	SURVEILLANCE	Schedule Follow-up	POST /api/audits/:id/phases/transition	closure certificate signed	buyer/auditor

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
Audit-Prep Agent (audit-agents)	POST /api/ai/audit-agents/prepare-questionnaire	PREP	Prep workspace · 'Generate questionnaire'	Risk-weighted questionnaire pulled from past findings + openFDA + EMA EudraGMDP + WHO PQ signals.
Supplier-Intel Agent (audit-agents)	POST /api/ai/audit-agents/supplier-intel	PREP PLANNING	Supplier register drawer	Public FDA warning letters + Form 483s + recalls for target supplier; verdict = known_tenant / public_only / ambiguous / unknown.
Auditor Coach Panel (Wave 3)	POST /api/ai/auditor-coach/suggest	EXECUTION	Auditor console side panel	Real-time observation severity guidance per ISO 19011 (Critical/Major/Minor/Observation).
Audit-Report Assembler (audit-agents)	POST /api/ai/audit-agents/assemble-report	FINDINGS CLOSURE	Findings drawer · 'Assemble report'	Generates findings log PDF + CAPA deadlines per GMP classification + SHA-256 integrity hash.

SCREENS

- /buyer/audits, /audits — Buyer / admin audit register
- /auditor/audits — Auditor workspace with phase kanban
- /audits/[id] — Audit detail with phase-specific artifact tabs

CLOSURE CONDITIONS

CLOSURE requires (a) all observations documented with severity + CAPA linkage, (b) closure certificate signed by buyer + supplier + auditor + witness roles, (c) all CAPAs closed. SURVEILLANCE triggers re-audit per risk-banded cadence.

10. Batch / Manufacturing Records

Compliance	21 CFR 211.188 · 21 CFR 211.192 · EU GMP Annex 11 (electronic batch records) · ICH Q7
Purpose	Capture the full manufacturing record for a single batch — BOM actuals, in-process tests, yield, equipment, linked deviations — then drive it through QA review to final disposition (release / reject / rework).
Model file	backend/src/models/BatchRecordModel.js
Collection	batch-records
Numbering	user-defined batchNumber

LIFECYCLE STATES

State	Primary owner	Description
MANUFACTURING	Operator	Batch in production; BOM + in-process tests + yields being recorded.
UNDER_REVIEW	Operator	Submitted for review; lab + deviation checks run.
PENDING_LAB_RESULTS	QC Lab	Waiting for analytical release results.
PENDING_QA_REVIEW	QA	Lab complete; QA reviewing batch record integrity.
PENDING_DEVIATION_CLOSURE	QA	Open deviations block release.
PENDING_DISPOSITION	VP / Director	Final release decision pending.
RELEASED	VP	Terminal. releaseDate set.
REJECTED	VP	Terminal. Batch destroyed or returned to supplier.
QUARANTINED	QA	Terminal (or non-terminal pending rework/reprocess).

LEGAL TRANSITIONS

From	To	UI button	API
(create)	MANUFACTURING	+ Create Batch	POST /api/batch-records

From	To	UI button	API
MANUFACTURING	UNDER_REVIEW PENDING_LAB_RESULTS PENDING_QA_REVIEW	<button>Submit for Review</button>	POST /api/batch-records/:id/submit review
UNDER_REVIEW PENDING_LAB_RESULTS PENDING_QA_REVIEW	PENDING_DEVIATION_CLOSURE PENDING_LAB_RESULTS PENDING_DISPOSITION	<button>QA Review</button>	POST /api/batch-records/:id/qa-review
PENDING_DISPOSITION	RELEASED	<button>Release</button>	POST /api/batch-records/:id/dispose (decision=RELEASED)
PENDING_DISPOSITION	REJECTED	<button>Reject</button>	POST /api/batch-records/:id/dispose (decision=REJECTED)
PENDING_DISPOSITION	QUARANTINED	<button>Quarantine / Rework</button>	POST /api/batch-records/:id/dispose (decision=REWORK REPROCESS QUARANTINE)

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
(roadmap) Yield anomaly detector	(not yet wired)	MANUFACTURING UNDER_REVIEW	(future)	Wave 3 roadmap — compare current yield + in-process test trends against historical and flag outliers for QA review.

SCREENS

- /batch-records — Batch register + lifecycle buttons (Submit / QA Review / Dispose)

CLOSURE CONDITIONS

RELEASED requires (a) batchRecord complete per approved MBR, (b) all in-process tests PASS, (c) all linked deviations CLOSED, (d) labResultsComplete=true, (e) QA reviewed, (f) VP disposition=RELEASED. releaseDate auto-set on RELEASED.

11. Equipment / Calibration

Compliance	ISO 9001:2015 §7.1.5 · 21 CFR 211.68(b) · ICH Q7 §5.3
Purpose	Track every GMP-relevant asset through its calibration lifecycle. Auto-escalate OVERDUE items. Tie failed calibrations to QUARANTINED status so the asset cannot be used in production.
Model file	backend/src/models/EquipmentModel.js
Collection	equipment
Numbering	EQ-YYYY-NNNN (auto)

LIFECYCLE STATES

State	Primary owner	Description
ACTIVE	Maintenance	In service, calibration current.
INACTIVE	Maintenance	Temporarily withdrawn, not under active maintenance.
UNDER_CALIBRATION	Maintenance	Currently being calibrated.
OUT_OF_SERVICE	Maintenance	Broken or awaiting repair.
QUARANTINED	QA	Failed calibration or suspected compromised; blocked from production use.
RETIRED	Maintenance	Terminal. Decommissioned; decommissionedAt set.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
(create)	ACTIVE	<button>+ Add Equipment</button>	POST /api/equipment	equipmentName, equipmentType, calibrationFrequencyDays (if requiresCalibration)	maintenance/admin
ACTIVE	UNDER_CALIBRATION	<button>Begin Calibration</button>	PUT /api/equipment/:id (status=UNDER_CALIBRATION)	—	maintenance
UNDER_CALIBRATION	ACTIVE (PASS) QUARANTINED (FAIL)	<button>Record Calibration</button>	POST /api/equipment/:id/calibration	performedAt, performedBy, result (PASS/CONDITIONAL/FAIL), certificateRef, nextDueDays	maintenance
ACTIVE	OUT_OF_SERVICE	<button>Mark Out of Service</button>	PUT /api/equipment/:id (status=OUT_OF_SERVICE)	—	maintenance

From	To	UI button	API	Required input	Allowed roles
QUARANTINED OUT_OF_SERVICE	ACTIVE	Return to Service	PUT /api/equipment/:id (status=ACTIVE) + new calibration	successful re-calibration	maintenance + QA
(any)	RETIRED	Retire	DELETE /api/equipment/:id (soft)	decommission reason	admin/tenant_admin

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
(roadmap) Predictive calibration	(not yet wired)	ACTIVE UNDER_CALIBRATION	(future)	Wave 3 roadmap — predict calibration failure probability from historical trend + usage hours + ambient conditions.

SCREENS

- /asset-management — Equipment register + calibration dialog (Record Calibration / Retire)

CLOSURE CONDITIONS

RETIRED is terminal. A QUARANTINED asset cannot be used in production until successfully re-calibrated (status flips back to ACTIVE with calibrationStatus=CURRENT).

12. Supplier Pre-Qualification

Compliance	ICH Q10 §2.7 · 21 CFR 211.84 (component qualification) · EU GMP Chapter 7
Purpose	Screen a new supplier before scheduling a full audit. Capture initial risk band, regulatory standards, product categories, and checklist compliance. Output: APPROVED / CONDITIONALLY_APPROVED / REJECTED.
Model file	backend/src/models/SupplierPreQualificationModel.js
Collection	supplier-pre-qualifications
Numbering	PQ-YYYY-NNNN (auto)

LIFECYCLE STATES

State	Primary owner	Description
DRAFT	Supplier	Supplier filling in pre-qual form.
SUBMITTED	Supplier	Submitted; buyer queue entry.
UNDER_REVIEW	Buyer / Auditor	Checklist being reviewed.
APPROVED	Auditor / VP	Terminal. Ready for full audit scheduling.
CONDITIONALLY_APPROVED	Auditor / VP	Terminal. Conditions captured + tracked separately.
REJECTED	Auditor / VP	Terminal.
EXPIRED	(auto)	Terminal. validUntil passed.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
(create)	DRAFT	+ Start Pre-Qual	POST /api/supplier-prequalifications	scope, regulatoryStandards[], productCategories[]	supplier/buyer/admin
DRAFT	SUBMITTED	Submit	PUT /api/supplier-prequalifications/:id (status=SUBMITTED)	full payload	supplier
SUBMITTED	UNDER_REVIEW	Begin Review	PUT /api/supplier-prequalifications/:id (status=UNDER_REVIEW)	checklist assigned	buyer
UNDER_REVIEW	APPROVED CONDITIONALLY_APPROVED REJECTED	Approve / Conditional / Reject	POST /api/supplier-prequalifications/:id/decision	decision, decisionNotes, validUntil, conditions[] (if conditional)	auditor/tenant_admin
APPROVED	(escalate to audit)	Schedule Full Audit	POST /api/audit-requests (from PQ)	audit scope	buyer
(any non-terminal)	EXPIRED	(auto)	(auto job)	validUntil passed	system

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
Supplier-Intel Agent (audit-agents)	POST /api/ai/audit-agents/supplier-intel	DRAFT UNDER_REVIEW	PQ detail · 'Check public signals'	Enriches the PQ record with openFDA + FDA warning letters + import alerts for the supplier.

SCREENS

- /supplier-prequalification — PQ register + decision drawer

CLOSURE CONDITIONS

Terminal states (APPROVED / CONDITIONALLY_APPROVED / REJECTED) require decision + decisionNotes. APPROVED sets validUntil (default 2 years).

13. Audit Request / RFQ (marketplace audit flow)

Compliance	ICH Q10 §2.7 · ISO 19011 §5.3 · contract audit governance
Purpose	Buyer posts an audit RFQ. Multiple auditor orgs submit quotes. Buyer shortlists + awards. Awarded quote converts into a full audit request that enters the 8-phase audit lifecycle.
Model file	backend/src/models/auditRfqModel.js · auditRfqQuoteModel.js
Collection	audit-rfqs + audit-rfq-quotes
Numbering	RFQ-XXXXXX (auto)

LIFECYCLE STATES

State	Primary owner	Description
DRAFT	Buyer	RFQ being drafted.
PUBLISHED	Buyer	Visible to invited auditors.
IN_QA	(internal)	Internal quality-check.
QUOTES_RECEIVED	Auditors	Auditors have submitted quotes.
SHORTLISTED	Buyer	Buyer has narrowed to top candidates.
AWARDED	Buyer	A quote has been accepted.
CONVERTED	(system)	Terminal. Awarded quote linked to an auditRequest row.
CANCELLED	Buyer	Terminal.
EXPIRED	(auto)	Terminal. closingAt passed without award.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
(create)	DRAFT	+ New RFQ	POST /api/rfqs	title, supplierOrgId, siteld, productIds[], closingAt	buyer
DRAFT	PUBLISHED	Publish	POST /api/rfqs/:id/publish	scope + closingAt	buyer
PUBLISHED	QUOTES_RECEIVED	(auto)	(auto when first quote submitted)	—	system
PUBLISHED	PUBLISHED (with invites)	Invite Auditors	POST /api/rfqs/:id/invite	auditorOrgIds[]	buyer
QUOTES_RECEIVED	SHORTLISTED	Shortlist	PUT /api/rfqs/:id (status=SHORTLISTED)	shortlist[]	buyer
QUOTES_RECEIVED SHORTLISTED	AWARDED	Award	POST /api/rfqs/:id/award (quoteId)	awardedQuoteld	buyer
AWARDED	CONVERTED	(auto)	(auto linked to POST /api/audit-requests)	audit request created	system
(auditor side)	(quote) SUBMITTED	Submit Quote	POST /api/rfqs/:id/quotes	lineItems[], totals, proposedSchedule	auditor

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
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SCREENS

- /rfqs — RFQ list (buyer)
- /rfqs/[id] — RFQ detail + quotes + Q&A threads + award
- /request-audit — Buyer audit-request creation

CLOSURE CONDITIONS

CONVERTED is the terminal happy-path state. The awarded auditRfqQuote binds to a newly-created audit request row, which then enters the 8-phase audit lifecycle (module 9).

14. Design Control (medical device)

Compliance	21 CFR 820.30 · ISO 13485:2016 §7.3 · ISO 14971 (risk mgmt)
Purpose	Track a medical-device design through 8 phases (Planning → Input → Output → Review → Verification → Validation → Transfer → Changes). Maintain the Design History File (DHF). Link to Risk, Change Control, CAPA.
Model file	backend/src/models/DesignControlModel.js
Collection	design-controls
Numbering	DC-YYYY-NNNN (auto)
Classification	CLASS_I CLASS_II CLASS_III IVD

LIFECYCLE STATES

State	Primary owner	Description
DRAFT	Product Engineer	Draft; editable before any phase starts.
ACTIVE	Product Engineer	At least one phase is IN_PROGRESS.
DESIGN_FREEZE	QA / RA	Design transfer complete; design frozen.
TRANSFERRED	Manufacturing	Formal design transfer to manufacturing site.
OBSOLETE	QA	Terminal. End-of-life.
CANCELLED	QA	Terminal.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
(create)	DRAFT	+ Start Design	POST /api/design-controls	title, productName, deviceClass, regulatoryPathway	product engineer/admin
DRAFT	ACTIVE	Activate	POST /api/design-controls/:id/advance-phase (to PLANNING)	first phase started	product engineer
ACTIVE	ACTIVE (next phase)	Advance Phase	POST /api/design-controls/:id/advance-phase	phase artifacts complete	product engineer/QA
ACTIVE (REVIEW+)	ACTIVE	Add Design Review	POST /api/design-controls/:id/reviews	reviewDate, attendees[], decision (PROCEED/REVISE/HOLD), actionItems[]	QA / reviewer
ACTIVE (TRANSFER)	DESIGN_FREEZE → TRANSFERRED	Transfer	POST /api/design-controls/:id/transfer	manufacturingSiteId	QA / RA / admin
TRANSFERRED	OBSOLETE	Obsolete	PUT /api/design-controls/:id (status=OBSOLETE)	end-of-life decision	QA
(any non-terminal)	CANCELLED	Cancel	PUT /api/design-controls/:id (status=CANCELLED)	cancellation reason	QA

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
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SCREENS

- /design-controls — Design register + phase tracker + review drawer

CLOSURE CONDITIONS

TRANSFERRED is the main happy-path terminal (device design released to manufacturing). OBSOLETE is the end-of-life terminal after market withdrawal.

Cross-cutting infrastructure

Electronic Signatures (21 CFR Part 11)

File: backend/src/routes/electronicSignatureRoutes.js

Endpoint: POST /api/electronic-signatures/sign

Generic e-signature endpoint accepts {recordType, recordId, signatureMeaning, authMethod}. Captures content SHA-256 hash, signer IP, user-agent. Signatures cannot be repudiated; logged to DataIntegrityLog. Used by document-control approvals, MRM completion, CAPA closure.

AI Audit Trail (FDA AI guidance · Jan 2025)

File: backend/src/services/ai/audit/aiAuditTrail.js

Endpoint: Internal: recordAiDecision() · POST /api/ai/decisions/outcome (user accept/reject)

Every AI decision logs feature, prompt-version, retrieval-set hashes, output, confidence, grounded flag, provider+model, latency, accept/reject outcome. Enables drift dashboard + regulator-ready evidence.

Universal Module Config

File: backend/src/middleware/universalPlatform.js

Endpoint: GET /api/universal-platform/config

Each tenant has modules.* boolean flags (EVENT_MANAGEMENT / DOCUMENT_CONTROL / RISK_MANAGEMENT / MANAGEMENT_REVIEW / CAPA / TRAINING / SUPPLIER_QUALITY / etc.). Frontend hides nav and 403s for disabled modules. Novex has all 15 enabled.

Role middleware

File: backend/src/middlewares/roleMiddleware.js

Endpoint: Used as middleware on every protected route

permit(...allowedRoles) gate. Roles normalised: tenantAdmin → tenant_admin, supplierUser → supplieruser, etc. Returns 403 with 'Forbidden: You don't have permission to access this resource.' if role not in allow-list.